

Control work 10, and work on the mistakes

Coordinates, functions and the straight line in one paper.

LESSON 143–144

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 10

S8 9–11 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Plot and read points accurately and name their quadrants.
- Evaluate a function and solve for the argument.
- Draw a linear graph and find its intercepts.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 407–431
Cambridge	Stage 8, pp. 88–116
Workbook	Control paper M10

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Plot $A(3; 2)$, $B(-4; 1)$, $C(-2; -3)$, $D(5; -1)$ and name each quadrant	4	L137– 138
2	Find the length and midpoint of the segment from $(1; 3)$ to $(7; 3)$	4	L137– 138
3	For $f(x) = 2x + 1$ find $f(3)$, $f(-4)$ and the x with $f(x) = 11$	4	L139– 140
4	Give the domain of $y = \frac{1}{x - 3}$	3	L139– 140
5	Draw $y = 2x - 4$ and mark both intercepts	5	L141– 142
6	Find the gradient through $(1; 2)$ and $(4; 8)$, and say whether $(3; 6)$ lies on that line	5	L141– 142

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for labelling the points and one for the quadrant names; Q3 two for the substitutions and two for solving the equation; Q5 two for the two intercepts and one for a ruled line; Q6 two for the gradient and two for the substitution check.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
coordinates reversed	$(3; 2)$	along first, then up
quadrants numbered clockwise	$(-4; 1)$	anticlockwise from the top right
midpoint subtracted	$7 - 1 = 6$	$\frac{1 + 7}{2} = 4$
negative substituted carelessly	$f(-4) = 9$	$2 \cdot (-4) + 1 = -7$
domain left blank	“every number”	$x \neq 3$
line drawn freehand	a wobbling curve	a ruled line through the intercepts
gradient differences out of order	$\frac{8 - 2}{1 - 4} = -2$	$\frac{8 - 2}{4 - 1} = 2$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

BLOCK	THE SENTENCE
the plane	two axes, one origin, and an ordered pair for every point
quadrants	numbered anticlockwise, with a sign pattern each
functions	one input, exactly one output
three descriptions	formula, table and graph say the same thing
the domain	the values of x the rule allows
the linear function	$y = kx + b$
the graph	a straight line, two points enough, third as a check

LOOKING FORWARD

Two lines that cross give a pair of numbers satisfying both equations at once. That is a **system**, and it is the next block of the quarter.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $f(x) = 2x + 1$.

① $f(3) = 7$

② $f(-4) = 2 \cdot (-4) + 1 = -7$

The bracket keeps the sign right.

③ $2x + 1 = 11$

④ $x = 5$

ANSWER

7, -7, $x = 5$

EXAMPLE 2 Model answer, Q5: draw $y = 2x - 4$.

① $x = 0$: $y = -4$, so $(0; -4)$.

② $y = 0$: $2x = 4$, so $(2; 0)$.

③ Rule a line through both.

④ Check with $x = 3$: $y = 2$ lies on it ✓

ANSWER

A line through $(0; -4)$ and $(2; 0)$

EXAMPLE 3 Model answer, Q6: gradient through $(1; 2)$ and $(4; 8)$, and is $(3; 6)$ on it?

$$① \quad k = \frac{8 - 2}{4 - 1} = 2$$

$$② \quad \text{Through } (1; 2): 2 = 2 \cdot 1 + b, \text{ so } b = 0.$$

$$③ \quad \text{The line is } y = 2x.$$

$$④ \quad 2 \cdot 3 = 6 \checkmark \text{ — the point lies on it.}$$

ANSWER $k = 2$; yes**03** Interactive model

Return Q5 first and ask who used a ruler; the mark for a straight line is the easiest in the paper and the most often lost.

The chapter in twelve questions

The first coordinate is read:

up the y -axis

along the x -axis

from the origin

either way

$(-2; -3)$ lies in quadrant:

I

II

III

IV

$(0; 4)$ lies:

in quadrant I

on the y -axis

on the x -axis

at the origin

The midpoint of $(1; 3)$ and $(7; 3)$ is:

$(4; 3)$

$(3; 3)$

$(6; 3)$

$(8; 3)$

A function gives each input:

one output

two outputs

many outputs

none

For $f(x) = 2x + 1$, $f(-4)$ is:

9

-7

-9

7

The domain of $y = \underline{1}x - 3$ is:

every number

$x \neq 0$

$x \neq 3$

$x > 3$

The graph of $y = kx + b$ is:

a curve

a straight line

two lines

a circle

$y = 2x - 4$ cuts Oy at:

$(0; -4)$

$(2; 0)$

$(0; 2)$

$(-4; 0)$

And cuts Ox at:

$(0; -4)$

$(2; 0)$

$(4; 0)$

$(-2; 0)$

The gradient through $(1; 2)$ and $(4; 8)$ is:

2

-2

3

6

Is $(3; 6)$ on $y = 2x$?

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Coordinate	Koordinata	Координата
Quadrant	Chorak	Четверть
Function	Funksiya	Функция
Domain	Aniqlanish sohasi	Область определения
Gradient	Burchak koeffitsiyenti	Угловой коэффициент
Intercept	Kesma	Свободный член
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 also asks for:

the lengths

the quadrants

the gradients

the domain

Q2 midpoint uses:

the difference

the average

the product

the sum

Q3 last part is:

a substitution

an equation

a graph

a domain

Q4 answer is:

every number

$x \neq 3$

$x \neq 0$

$x > 0$

Q5 is drawn with:

freehand

a ruler

compasses

a protractor

Q6 checks the point by:

plotting only

substituting

measuring

guessing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The quadrant of $(3; 2)$

ANSWER I

2. The quadrant of $(-2; -3)$

ANSWER III

3. The midpoint of $(1; 3)$ and $(7; 3)$

ANSWER $(4; 3)$

4. $f(x) = 2x + 1$: $f(3)$

ANSWER 7

5. $f(-4)$

ANSWER -7

6. The domain of $y = \frac{1}{x - 3}$

ANSWER $x \neq 3$

7. $y = 2x - 4$ cuts Oy at

ANSWER $(0; -4)$

Medium · the standard question

7 PROBLEMS

1. $f(x) = 2x + 1 = 11$

ANSWER $x = 5$

2. $y = 2x - 4$ cuts Ox at

ANSWER $(2; 0)$

3. The gradient through $(1; 2)$ and $(4; 8)$

ANSWER 2

4. Is $(3; 6)$ on $y = 2x$?

ANSWER Yes

5. The length from $(1; 3)$ to $(7; 3)$

ANSWER 6

6. The equation of the line through $(1; 2)$ with gradient 2

ANSWER $y = 2x$

7. Where is $(6; 0)$?

ANSWER On the x -axis

Hard · stretch and reason

7 PROBLEMS

1. Where do $y = 2x - 4$ and $y = -x + 5$ cross?

ANSWER $(3; 2)$

2. A line through $(0; -4)$ and $(2; 0)$: its equation

ANSWER $y = 2x - 4$

3. For which k does $y = kx - 4$ pass through $(1; -1)$?

ANSWER $k = 3$

4. The domain of $y = \frac{1}{x^2 - 9}$

ANSWER $x \neq \pm 3$

5. The fourth vertex of a rectangle on $(1; 1)$, $(5; 1)$, $(5; 4)$

ANSWER $(1; 4)$

6. $f(x) = 2x + 1$: for which x is $f(x) = x$?

ANSWER $x = -1$

7. Two lines with equal gradients are

ANSWER parallel

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Plot $(4; 3)$, $(-1; 5)$, $(-3; -2)$ and $(2; -4)$ and name the quadrants.
3. For $f(x) = 3x - 2$ find $f(4)$, $f(-2)$ and the x with $f(x) = 10$.
4. Draw $y = -2x + 6$ and mark both intercepts.

5. Find the gradient of the line through $(-1; 1)$ and $(3; 9)$.